

Me Versus Everyone: Discrepancies Between General and Personal Emotion Malleability Beliefs Predict Clinical Symptoms, Affect, and Emotion Regulation

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One factor that relates to clinical symptoms, affect, and emotion regulation is beliefs that individuals hold about the nature of emotions, specifically the degree to which they view emotions as changeable (E. T. Kneeland, Dovidio, et al., 2016). Across two studies (Study 1, $N = 371$, data collected from June 2021 to May 2023; Study 2, $N = 143$, data collected from January 2023 to December 2023), we examined how the *discrepancy* between individuals' beliefs about the malleability of emotion in general relative to their own emotions related to and predicted clinical symptoms, affect, and emotion regulation efforts. Mechanisms linking the discrepancy in emotion beliefs to emotional distress and emotion regulation also were investigated, as well as how differences in emotion beliefs varied by depression status (Study 2). In line with hypotheses, a stronger bias toward viewing emotion in general as more malleable compared to one's own emotions was associated with higher psychological distress and less active emotion regulation concurrently and longitudinally. As expected, individuals with past depression or no history of depression had more of a bias toward viewing their own emotions as more malleable compared to emotion in general. The present studies address a gap in existing emotion belief research to clarify how discrepancies in general versus personal emotion beliefs are associated with emotional distress and emotion regulation and mechanisms in these relationships.

Keywords: emotion, emotion beliefs, emotion regulation, depression

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Individuals employ a range of strategies to manage their experience and expression of emotion—referred to as emotion regulation (Gross, 1999, 2015). Importantly, differences in the emotion regulation strategies individuals select and employ are associated with psychological distress, including depression, anxiety, and substance use (Aldao et al., 2010; Sheppes et al., 2015). One factor that relates to and can influence emotion regulation and emotional distress is the beliefs that individuals hold about the nature of emotions—emotion beliefs (Ford & Gross, 2019; E. T. Kneeland, Dovidio, et al., 2016). One type of emotion belief that individuals can hold centers on the degree to which they believe that emotions can change—also called emotion malleability beliefs (E. T. Kneeland, Dovidio, et al., 2016).

Research on emotion malleability beliefs draws directly from Carol Dweck's robust research on the role of implicit theories of intelligence, which centers on the degree to which individuals believe that a construct, such as intelligence, is changeable versus fixed (Blackwell et al., 2007; Dweck & Leggett, 1988). This work clarified that implicit theories directly influence effort expended at overcoming a related challenge,

feelings of helplessness and stress, and ultimately perceptions of agency (Blackwell et al., 2007; Dweck et al., 1995; Hong et al., 1999; Lee et al., 2019). More recently, Dweck's work on implicit theories has been extended to the domain of emotion (Ford & Gross, 2019; E. T. Kneeland, Dovidio, et al., 2016; Tamir et al., 2007). Implicit theories of emotion—also called emotion malleability beliefs—have been correlated with levels of depression, anxiety, and negative affect (E. T. Kneeland, Dovidio, et al., 2016; Schroder et al., 2015; Tamir et al., 2007). In terms of emotion regulation, more malleable views of emotions in general are positively correlated with the greater use of cognitive reappraisal, an active emotion regulation strategy that involves reframing an emotional situation in a new light, and decreased avoidance (Kappes & Schikowski, 2013; E. T. Kneeland, Nolen-Hoeksema, et al., 2016a, 2016b; Tamir et al., 2007). Experimental and longitudinal work have also demonstrated that more malleable emotion beliefs about emotions on a general level predict lower depression, greater cognitive reappraisal, and less rumination (E. T. Kneeland, Dovidio, et al., 2016; E. T. Kneeland & Dovidio, 2020; Tamir et al., 2007). Moreover, differential

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engagement in emotion regulation, specifically cognitive reappraisal and rumination, represents the pathway linking more malleable general emotion beliefs to lower emotional distress (Gutentag et al., 2017; E. T. Kneeland & Dovidio, 2020; Tamir et al., 2007).

While much extant work on emotion malleability beliefs has focused on the malleability of emotions in general, individuals can also hold distinct beliefs about the nature of their *own* emotions. De Castella et al. (2013) clarified that more malleable views of one's own emotions were associated with decreased psychological distress, specifically depression and stress, and increased use of cognitive reappraisal. Importantly, although general and personal emotion malleability beliefs were correlated, personal emotion malleability beliefs had stronger relationships with these outcomes when directly compared to malleability beliefs about emotions in general (De Castella et al., 2013). De Castella and Byrne (2015) also clarified that personal beliefs are better predictors of outcomes such as helplessness and lower motivation. Taken together, this logic suggests that individuals can hold different beliefs regarding the malleability of emotions in general and their own emotions and these two different sets of emotion beliefs can differentially relate to clinical symptoms and emotion regulation.

Empirical evidence demonstrates that there can exist discordance across different measures and aspects of emotion. Individuals can experience discrepancies between physiological experiences of emotion and their own subjective reports of emotional experiences, with these measures often showing small to absent correlations, reflecting relatively low concordance (Friedman et al., 2014). For example, De Los Reyes and colleagues found that self-reported measures of social anxiety demonstrated low concordance with physiological assessment of anxiety (De Los Reyes et al., 2012). There can also exist differences in an individual's report of their own emotional experiences compared to how a close social other perceives the target individual's emotions. For example, accuracy of emotion perception can vary based on the specific emotion being expressed and the partner's own current emotion state (Clark et al., 2017). Therefore, aspects of one's internal emotional experiences can be discordant as well as reports about an individual's emotion states by a close social other. Additionally, individuals can hold different beliefs about the affect they would like to experience—ideal affect—and their current affective state (Tsai et al., 2006). These authors also found that a greater discrepancy one perceives between their current and ideal affect is associated with greater depression. Taken together, there can exist discordance between measures of emotion within an individual, as well as differences between self and other perceptions of an individual's emotion.

To our knowledge, it remains an open empirical question about how discrepancies between the beliefs that individuals hold about emotions *in general* compared to *their own* emotions then relate to emotion regulation, affect, and clinical symptoms. In the realm of emotion beliefs, there can be divergence in terms of the benefits of believing that emotions are changeable based on context. As described earlier, there exist benefits for psychological health related to more changeable views of emotion. Yet conversely, when participants were presented with a hypothetical case of a depressed individual, stronger views that the individual's emotions were changeable were associated with more avoidance and less support (Smith et al., 2023). Therefore, the same type of emotion belief can hold different outcomes based on whether the belief centers on a personal attribute or that same attribute in another individual.

The present study addresses this empirical gap in the emotion belief literature to investigate how the magnitude of a discrepancy between individuals' beliefs about the malleability of *their own* emotions and emotions *in general* relates to their level of clinical symptoms and emotion regulation. Beliefs about other's emotions may better reflect a person's view of norms about emotions, which can guide the regulation and expression of emotion (Eid & Diener, 2001) and are seen as a reflection of the general consensus around how emotions are experienced or expressed in a particular society (Vishkin & Tamir, 2023). Research on emotion norms proposes that the greater the degree to which one perceives that their emotions vary from a more general emotion norm, this perceived discrepancy increases negative affect (Bastian et al., 2012; Vishkin & Tamir, 2023). For example, Bastian et al. (2012) found that when individuals have a greater perception that others think they should not be feeling a negative emotion, like sadness, this increases the individual's negative emotion. By contrast, one's personal emotion beliefs may reflect their own experiences with emotional challenges and not necessarily match more general emotion norms. Thus, larger discrepancies between beliefs about the malleability of emotions in general and one's own emotions may reflect an individual's perceived failure to experience emotions in a "normative" manner.

In terms of levels of depression and anxiety, having a stronger view that emotions in general are more malleable than your own could feed emotional distress. Self-discrepancy theory proposes that greater discrepancies between aspects of the self, such as between our actual self and ideal self, can feed psychopathology (Mason et al., 2019). By extension, it could be that holding different beliefs about the nature of your own emotions compared to others could similarly increase negative emotions. From an emotion regulation perspective, holding more discrepant views between the malleability of one's own emotions and emotions in general could feed a sense of helplessness regarding changing one's own emotions (E. T. Kneeland, Dovidio, et al., 2016). This could incline individuals away from active emotion regulation strategies, such as cognitive reappraisal, and more toward passive emotion regulation strategies, such as rumination. This difference in emotion regulation tied to this emotion belief discrepancy could, in turn, increase and maintain psychological distress (Aldao et al., 2010). Relatedly, the view that one's own emotions are more malleable than emotions in general could incline individuals to engage in more active emotion regulation efforts, because they believe that their own emotions are entities that are subject to change through their own active effort. This stance that one's own emotions are more malleable than emotions in general could also similarly lead to lower levels of depression and anxiety stemming from the view that one's own emotions are changeable.

The goal of the present study is to address a gap in the emotion belief literature to examine differences in the beliefs that individuals hold about the malleability of emotions, in general, and their own emotions and how this discrepancy is associated with clinical symptoms, affect, and emotion regulation tendencies. We hypothesized that a greater discrepancy between general and personal emotion beliefs in the direction of holding more malleable views of emotion in general compared to one's own emotion would be associated with (a) higher levels of depression, anxiety, and rumination and (b) lower levels of cognitive reappraisal. Additionally, we hypothesized that these relationships would also occur across time with a greater discrepancy in emotion malleability beliefs predicting higher clinical symptoms and less active emotion regulation at 1-month follow-up. For the

relationships that were significant longitudinally, we conducted follow-up analyses to clarify whether the path linking discrepancies in emotion beliefs would relate to clinical symptoms via differential engagement in emotion regulation, in line with past research (E. T. Kneeland & Dovidio, 2020; Tamir et al., 2007). Finally, on an exploratory basis, we examined whether levels of discrepancy between general and personal emotion beliefs would vary based on depression status (current, remitted, or no history). We hypothesized that individuals in a current depressive episode would have the greatest bias toward thinking that emotions in general were more malleable than their own, followed by individuals with a past history of depression.

Relationships between differences in emotion beliefs, clinical symptoms, affect, and emotion regulation were examined across two samples—undergraduate students (Study 1) and adults drawn from the community (Study 2)—and were examined both concurrently and longitudinally at 1 month after baseline. In Study 2, differences in these associations were also examined based on depression status (current depression, past depression, no history of depression) as assessed using a semistructured clinical interview. The current investigation represents the first step in providing evidence that a discrepancy between beliefs about the malleability of *one's own* emotions and emotions *in general* is an important factor that relates to and predicts clinical symptoms, levels of affect, and emotion regulation.

Study 1

Method

Participants

Participants were recruited from a small liberal arts college in the Northeast United States and completed an in-person lab session as part of a larger study on the emotion beliefs and mental health of undergraduates in college. Qualtrics' secure online survey platform was utilized to collect all data. The sample consisted of 371 individuals: 218 women (58.8%), 133 men (35.8%), and 18 nonbinary or transgender individuals (4.9%). Two individuals (.5%) chose not to report their gender. The sample included 171 White (46.1%), 38 Black or African American (10.2%), 94 Asian or Asian American (25.3%), 49 multiracial (13.2%), three Native American or Alaskan Native (.8%), one Native Hawaiian or Pacific Islander (.3%), and two Caribbean Islander (.5%) individuals. Fourteen (3.8%) participants identified as Hispanic or Latinx. The average age of the sample was 19.62 years ($SD = 1.44$, range = 17–28), and 108 participants (29.0%) identified as part of the lesbian, gay, bisexual, transgender, queer+ community. A sensitivity analysis was conducted using the G*Power software program after the conclusion of data collection based on an α level of .05 and with 90% power. This analysis determined that the current sample size had the ability to detect effects larger than .027. There were no significant differences in key study variables between participants who did and did not complete the 1-month follow-up survey ($ps > .068$).

Procedure

Study approval was obtained through the affiliated Institutional Review Board. Participants were recruited as part of a larger multipurpose study, and the present study's sample size was determined based on sample sizes of similar study designs (e.g., Brown et al., 2021). All participants gave informed consent and were told that the goal of the study was to improve the mental health of college students.

Participants completed a demographics questionnaire and additional questionnaires measuring trait beliefs about the malleability of emotions, emotion regulation, and clinical symptoms as part of a larger multipurpose study. Data were collected from June 2021 to May 2023. All participants passed study attention checks and the minimum cutoff for time spent taking the survey. Participants were debriefed of the study's purpose and provided with course credit or \$15 as compensation.

Measures

The Implicit Theories of Emotion Scale (ITES; Tamir et al., 2007; $\alpha = .710$, $\omega = .717$) is a four-item self-report scale that measures general beliefs about the malleability and controllability of emotions. The scale is rated on a 6-point scale, with "1" being *strongly disagree* and "6" being *strongly agree*. The ITES includes items like "No matter how hard I try, I can't really change the emotions that I have." A modified version of the ITES assesses personal beliefs about how malleable and controllable one's own emotions are (De Castella et al., 2013; $\alpha = .815$, $\omega = .818$) and includes items like "The truth is, I have very little control over my emotions." For both scales, lower scores indicate more fixed views of emotions, while higher scores indicate more malleable views of emotions.

To measure cognitive reappraisal use, participants completed the Emotion Regulation Questionnaire (ERQ; Gross & John, 2003; $\alpha = .743$, $\omega = .744$) reappraisal subscale. The ERQ reappraisal subscale is a six-item self-report scale that is rated on a 7-point Likert scale ranging from *strongly disagree* to *strongly agree*. The ERQ reappraisal subscale includes items like "When I want to feel less negative emotion, I change the way I'm thinking about the situation." Higher scores indicate a greater use of cognitive reappraisal, while lower scores indicate less frequent use. The expressive suppression scale was originally included in earlier waves of the study; the present studies include only the reappraisal subscale because this subscale was assessed across all waves of data collection and given the consistent ties between emotion malleability beliefs and the state and trait use of cognitive reappraisal (e.g., E. T. Kneeland, Nolen-Hoeksema, et al., 2016a, 2016b; Tamir et al., 2007).

The Brooding Subscale Ruminative Response Scale (RRS; Nolen-Hoeksema et al., 1994; $\alpha = .762$, $\omega = .768$) is a five-item self-report scale that measures how frequently rumination is used as an emotion regulation strategy. For example, participants were asked to indicate how often they think "why can't I handle things better?" The scale is rated on a 4-point Likert scale ranging from *never* to *almost always*. Higher scores indicate a greater tendency to ruminate.

To measure trait levels of affect over the last 2 weeks, all participants completed the 10-item Positive and Negative Affect Schedule (PANAS; Watson et al., 1988; PA: $\alpha = .855$, $\omega = .856$; NA: $\alpha = .850$, $\omega = .854$). Examples of items on the PANAS include "Scared" and "Determined." Items are on a 5-point Likert scale, with 1 being *very slightly or not at all* and 5 being *extremely*. Higher scores on the negative affect subscale indicate higher levels of negative affect, and higher scores on the positive affect subscale indicate higher levels of positive affect.

The Beck Depression Inventory (BDI-II; Beck et al., 1996; $\alpha = .902$, $\omega = .905$) assessed trait levels of depression in the last 2 weeks. Each item is rated on a 4-point scale, with higher scores indicating greater depression severity. The Perceived Stress Scale (PSS; Cohen et al., 1983; $\alpha = .868$, $\omega = .874$) assessed trait levels of stress in the

past month. Each item is rated on a 5-point scale, with higher scores indicating higher levels of perceived stress. The PSS includes items like “In the last month, how often have you been upset because of something that happened unexpectedly?” with “0” being *never* and “4” being *very often*.

1-Month Follow-Up Survey

Participants that indicated they would be interested in follow-up surveys were sent a 1-month follow-up survey that assessed depression, stress, emotion regulation, and self-efficacy. A total of 105 participants completed the 1-month follow-up survey, during which they completed the BDI-II (Beck et al., 1996; $\alpha = .909$, $\omega = .912$), ERQ (Gross & John, 2003; $\alpha = .887$, $\omega = .885$), PANAS (Watson et al., 1988; PA: $\alpha = .908$, $\omega = .911$; NA: $\alpha = .838$, $\omega = .844$), and RRS (Nolen-Hoeksema et al., 1994; $\alpha = .856$, $\omega = .863$). Upon completing the survey, participants were debriefed about the study’s purpose and were compensated \$5 for their participation.

Statistical Analyses

IBM SPSS Statistics 27 was utilized to analyze all data. The difference score between general and personal emotion malleability beliefs was calculated by subtracting personal emotion beliefs from general emotion beliefs; therefore, higher positive scores indicated that individuals had more malleable views of emotions in general compared to their own emotions, and more negative scores indicated the opposite—that participants held more malleable views of their own emotions compared to emotions in general. Bivariate correlations were conducted initially to determine whether there were significant associations with the difference between participants’ personal and general emotion beliefs and clinical symptoms, affect, and emotion regulation. For longitudinal analyses, when the initial correlation was significant, follow-up linear regressions were conducted with the baseline version of the 1-month variable (e.g., baseline BDI when 1-month BDI was the dependent variable) as an additional predictor in the model.

Follow-up models were conducted to examine pathways linking differences in emotion beliefs to main study outcomes using the PROCESS Macro for SPSS (Model 4, 95% CI, 5,000 bootstrapped samples; Hayes, 2022).

Transparency and Openness

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study. All measures and manipulations that were included in both studies as part of a larger multipurpose study are listed in the [Supplemental Material](#). All data and study materials are available at https://osf.io/w4pny/?view_only=a6f10ee7132a4825a22b524e3cc489ba. The studies’ designs and hypotheses were not preregistered.

Results

Table 1 presents descriptive statistics for all main study variables. Table 2 presents correlations between differences in beliefs about the malleability of general versus personal emotions and clinical symptoms, affect, and emotion regulation. Overall, individuals held more malleable views of their own emotions ($M = 3.61$, $SD = 0.75$)

compared to emotions in general ($M = 3.36$, $SD = 0.71$). In line with hypotheses, more of a difference in emotion beliefs in the direction of believing that emotions in general are more malleable than your own emotions was associated with higher levels of depression, stress, and rumination and lower levels of reappraisal and trait positive affect. There was not a significant correlation between differences in emotion beliefs and trait negative affect ($p = .357$). We also examined models with both general and personal emotion malleability beliefs as predictors in analyses. For parsimony, we focused only on differences between emotion beliefs as the independent variable in the main text. Information on results from analyses with personal emotion beliefs only is presented in the [Supplemental Material](#).

We next examined these relationships longitudinally between baseline differences in general and personal emotion malleability beliefs and clinical symptoms, affect, and emotion regulation as assessed 1 month after the baseline assessment. In line with hypotheses, more of a difference between general and personal emotion beliefs was negatively associated with the use of cognitive reappraisal ($R^2 = .084$, standardized $B = -.289$, $SE = 1.15$, $p = .003$, 95% CI $[-5.80, -1.23]$) in that more malleable beliefs about one’s own emotions compared to emotions in general were associated with the greater use of cognitive reappraisal. Importantly, using a multiple linear regression, this relationship between differences in emotion malleability beliefs at baseline

Table 1
Means and Standard Deviations for Study 1 and Study 2 Main Variables

Dependent variable	<i>M</i>	<i>SD</i>
Study 1		
General emotion malleability beliefs	3.36	0.70
Personal emotion malleability beliefs	3.61	0.75
Emotion belief discrepancy	−0.25	0.64
Depression symptoms	12.58	9.64
Perceived stress	18.18	6.71
ERQ reappraisal	30.53	5.40
RRS rumination	12.08	3.31
Positive affect	15.64	3.74
Negative affect	12.35	4.56
1M depression symptoms	0.97	8.81
1M perceived stress	13.46	4.28
1M emotion belief discrepancy	28.12	6.51
1M rumination	10.54	3.82
1M positive affect	22.60	7.35
1M negative affect	13.74	4.89
Study 2		
General emotion malleability beliefs	3.42	0.70
Personal emotion malleability beliefs	3.64	0.81
Emotion belief discrepancy	−0.23	0.58
Depression symptoms	11.25	10.25
Perceived stress	17.42	6.59
ERQ reappraisal	30.20	7.05
RRS rumination	11.80	3.28
Positive affect	14.49	4.37
Negative affect	11.13	4.29
1M depression symptoms	12.52	11.36
1M perceived stress	17.02	6.67
1M emotion belief discrepancy	27.77	7.05
1M rumination	10.38	3.88
1M positive affect	23.83	7.08
1M negative affect	13.39	5.49

Note. ERQ = Emotion Regulation Questionnaire; RRS = Ruminative Response Scale; 1M = 1-month follow-up survey variables.

Table 2
Correlations Between Differences in Emotion Beliefs, Clinical Symptoms, Affect, and Emotion Regulation Assessed at Baseline and 1-Month Follow-Up in Study 1

Outcome variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. ITES general emotion beliefs	—	.65**	.36**	-.20**	-.20**	.18**	-.11*	-.26**	.25**	-.19*	-.27**	.11	-.14	.11	-.13
2. ITES personal emotion beliefs		—	-.48**	-.34**	-.33**	.33**	-.24**	.46**	-.30**	-.20*	-.27**	.31**	-.09	.17	-.15
3. ITES baseline discrepancy			—	.20**	.17**	-.19**	.16**	-.29**	.08	.02	.01	-.29**	-.07	-.10	.03
4. BDI baseline score				—	.68**	-.31**	.47**	-.62**	.60**	.77**	.60**	-.42**	.60**	-.43**	.51**
5. PSS baseline score					—	-.25**	.49**	-.61**	.60**	.59**	.69**	-.36**	.55**	-.41**	.53**
6. ERQ baseline reappraisal score						—	-.27**	.29**	-.07	-.37**	-.43**	.50**	-.23*	.41**	-.32**
7. Brooding RRS trait score							—	-.21*	.17	.46**	.42	-.22*	.63**	-.26**	.40**
8. PANAS PA trait score								—	—	—	—	—	—	—	—
9. PANAS NA trait score									—	—	—	—	—	—	—
10. IM BDI total score										—	.70**	-.39**	.68**	-.58**	.72**
11. IM PSS total score											—	-.41**	.64**	-.59**	.59**
12. IM ERQ total score												—	-.29**	.48**	-.29**
13. IM RRS total score													—	-.45**	.66**
14. IM PANAS PA score														—	-.34**
15. IM PANAS NA score															—

Note. The ITES baseline discrepancy variable refers to the general–personal emotion malleability beliefs at baseline. Higher scores on this discrepancy variable refer to stronger beliefs that emotions in general are more malleable compared to one's own emotions. ITES = Implicit Theories of Emotion Scale; BDI = Beck Depression Inventory; PSS = Perceived Stress Scale; ERQ = Emotion Regulation Questionnaire; RRS = Ruminative Response Scale; PANAS = Positive and Negative Affect Schedule; PA = positive affect; NA = negative affect; IM = 1-month follow-up survey variables. * $p < .05$. ** $p < .01$. *** $p < .001$.

and cognitive reappraisal at 1 month remained significant when baseline levels of cognitive reappraisal were entered as an additional predictor in a linear regression model ($R^2 = .304$, standardized $B = -.233$, $SE = 1.02$, $p = .006$, 95% CI $[-4.85, -.82]$). Contrary to expectations, there were no significant relationships between baseline differences in emotion malleability beliefs and clinical symptoms, affect, and rumination as assessed at the 1-month follow-up. Assumptions for multiple regressions were met.

We next conducted a series of mediation analyses to examine whether differences in depression or positive affect represented an indirect path linking the significant relationship between differences in emotion beliefs at baseline and cognitive reappraisal at 1 month, given correlations between these variables at baseline. The indirect path linking differences in emotion beliefs to 1-month cognitive reappraisal via 1-month depression (indirect effect = -0.10 , bootstrap $SE = 0.45$, 95% CI $[-1.02, 0.81]$) and via 1-month positive affect (indirect effect = -0.53 , bootstrap $SE = 0.52$, 95% CI $[-1.48, 0.61]$) was not significant.

Study 2

The main aim of Study 2 was to extend the results of Study 1 to adults drawn from the surrounding communities. Analyses paralleled Study 1 in that we first examined relationships between emotion belief discrepancies and emotional distress (affect and clinical symptoms) and emotion regulation at baseline. We then examined these relationships across time using data from the 1-month follow-up. When longitudinal relationships between emotion belief discrepancies and emotional distress were significant, we conducted secondary mediation analyses to clarify whether differences in emotion regulation represented the indirect path in these relationships. When the longitudinal association between baseline emotion belief discrepancies and emotion regulation at 1 month was significant, we conducted a parallel set of analyses to clarify whether differences in emotional distress represented the link in these relationships. Past theoretical research proposed that differences in emotional distress, such as levels of anxiety symptoms, can mediate the relationship between emotion malleability beliefs and emotion regulation (E. T. Kneeland, Dovidio, et al., 2016). Past empirical research has found links between more fixed beliefs about emotion and both greater depressive symptoms and the greater use of rumination (E. T. Kneeland et al., 2020; E. T. Kneeland & Dovidio, 2020). More recent research also found that differences in emotional distress mediated the relationship between personal emotion beliefs and emotion regulation (E. T. Kneeland et al., 2024). Therefore, it is also important to examine the role of differences in emotional distress as a potential link between emotion beliefs and emotion regulation strategies. In Study 2, we also collected diagnostic information on participants' current depression status via a semistructured clinical interview that we then used to compare emotion belief discrepancies and these relationships between groups in a series of exploratory analyses.

Method

Participants

Participants were recruited from the surrounding community via flyers or social media as part of a larger study on emotions and mental health. All potential participants ($N = 177$) completed a phone screen to determine whether they met the exclusion criteria of a current

psychotic disorder or current severe substance use disorder or were not able to provide informed consent based on significant cognitive impairment. Fourteen participants were excluded due to meeting criteria for a diagnosis of current psychotic symptoms, and three were excluded due to meeting criteria for a diagnosis of a current severe substance use disorder. A total of 143 participants were included in the final analyzed sample: 98 women (68.5%), 22 men (15.4%), 19 nonbinary or transgender individuals (13.3%), and four participants who did not report gender (2.8%). Ninety White (62.9%), 27 Asian or Asian American (18.9%), 20 multiracial (14%), one Black or African American (.7%), and one Caribbean Islander (.7%) self-reported their ethnicities. Four participants did not report ethnicity (2.8%), and eight (5.6%) participants also identified as Hispanic or Latinx. A total of 58 (40.6%) participants reported being part of the lesbian, gay, bisexual, transgender, queer+ community. The average age of the sample was 27.64 years ($SD = 14.06$, range = 18–96 years). A sensitivity analysis was conducted using the G*Power software program after the conclusion of data collection based on an α level of .05 and with 90% power. This analysis determined that the current sample size had the ability to detect effects larger than .070. There were no significant differences in key study variables between participants who did and did not complete the 1-month follow-up survey ($ps > .051$).

Procedure

Study procedures were approved by the affiliated Institutional Review Board. Sample size was determined based on a previous study of similar design (E. T. Kneeland et al., 2020). Data were collected from January 2023 to December 2023. All participants were informed that the study's purpose was to help improve and find interventions to improve the mental health of adults in the community. For the pre-screening interview, the Structured Clinical Interview for *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition, Research Version (SCID-RV; First et al., 2015) Module B for Psychosis and Associated Symptoms and SCID-RV Module E for Substance Use Disorders were administered over the phone by trained researchers, as well as the Telephone Interview for Cognitive Status (Brandt et al., 1988) for participants over 65. Those that passed the phone screening ($N = 160$) underwent diagnostic interviews by a member of the research team (MS) who has a bachelor's degree in psychology and who was trained to reliability (100% diagnostic agreement) using the SCID Modules for mood and anxiety disorders, under supervision of the first author (EK). Participants ($N = 143$) that completed the baseline survey were included in the final analyzed sample. Based on the SCID, 80 participants met criteria for past major depressive disorder (MDD), 18 participants met criteria for current MDD, and 45 participants received no depression diagnosis.

All survey data were collected on Qualtrics' secure survey platform. Participants completed questionnaires measuring trait emotion beliefs of malleability of emotions and specific facts of emotions, their trait levels of perceived stress, and trait levels of depression. At the end of the study, all participants were then debriefed about the study's purpose and procedure and were provided with mental health resources. Participants were compensated \$15 for completing the SCID interview and \$15 for completing the survey.

Measures

As part of the baseline survey, participants filled out demographic information and a series of questionnaires measuring their beliefs about emotions, emotion regulation tendencies, and clinical symptoms. All participants completed the ITES general ($\alpha = .724$, $\omega = .736$) and personal malleability beliefs ($\alpha = .877$, $\omega = .881$; De Castella et al., 2013; Tamir et al., 2007). Participants completed the ERQ (Gross & John, 2003; $\alpha = .874$, $\omega = .868$) to measure cognitive reappraisal use. The RRS (Nolen-Hoeksema et al., 1994; $\alpha = .788$, $\omega = .796$) measured how frequently individuals used rumination as an emotion regulation strategy. The shortened 10-item version of the PANAS assessed current levels of positive and negative affect (Watson et al., 1988; PA: $\alpha = .786$, $\omega = .810$; NA: $\alpha = .840$, $\omega = .831$). The BDI-II (Beck et al., 1996) was used to measure trait levels of depression in the last 2 weeks ($\alpha = .933$, $\omega = .936$), and the PSS (Cohen et al., 1983; $\alpha = .855$, $\omega = .863$) assessed trait levels of perceived stress in the past month.

1-Month Follow-Up

Participants that indicated they would be interested in follow-up surveys were sent a 1-month follow-up survey 1 month after they completed their baseline survey that assessed depression, stress, and emotion regulation. A total of 114 participants completed the 1-month follow-up survey, during which they once again completed the BDI-II (Beck et al., 1996; $\alpha = .943$, $\omega = .946$), the PSS (Cohen et al., 1983; $\alpha = .869$, $\omega = .875$), and to assess cognitive reappraisal and rumination, the ERQ (Gross & John, 2003; $\alpha = .927$, $\omega = .930$) and the RRS, respectively (Nolen-Hoeksema et al., 1994; $\alpha = .879$, $\omega = .885$). To assess levels of affect, participants completed the positive ($\alpha = .873$, $\omega = .876$) and negative affect ($\alpha = .883$, $\omega = .886$) subscales of the PANAS (Watson et al., 1988). After completing the 1-month follow-up survey, participants were debriefed about the study's purpose and were compensated \$5.

Statistical Analyses

IBM SPSS Statistics 27 was used to analyze all data. Differences between general and personal emotion beliefs were calculated using the same procedure as in Study 1—more positive scores indicated a stronger sense that emotions in general were more malleable than one's own emotions. Following the same data analytic procedures used in Study 1, initial bivariate correlations were conducted to investigate significant associations between the ITES difference score and specific emotion regulation strategies and clinical symptoms at baseline. Bivariate correlations were also conducted to examine associations between differences in emotion beliefs and the same variables as assessed during the 1-month survey. When the initial correlation was significant, follow-up linear regressions were conducted with the baseline version of the 1-month variable (e.g., baseline BDI when 1-month BDI was the dependent variable) as an additional predictor in the model.

We next conducted a series of analyses using the PROCESS Macro in SPSS (Model 4, 5,000 bootstraps; Hayes, 2013) to clarify whether emotion regulation as assessed at the 1-month follow-up represented the mechanism linking differences in baseline emotion beliefs to clinical symptoms and affect at the 1-month follow-up. We also

conducted a parallel set of analyses to examine whether differences in levels of emotional distress (affect or clinical symptoms) represented an indirect path tying baseline emotion belief discrepancies to differences in emotion regulation efforts as measured at the 1-month follow-up. To examine whether differences in emotion beliefs varied based on depression status, an analysis of variance was conducted with depression status (current MDD, past MDD, or no MDD) as the fixed factor and emotion belief differences as the dependent variable. Follow-up analyses examined bivariate correlations between emotion belief differences and main study outcomes (e.g., clinical symptoms, affect, and emotion regulation) based on depression group status.

Results

Overall, paralleling Study 1, participants held more malleable views of their personal emotions ($M = 3.64$, $SD = 0.81$) compared to emotions in general ($M = 3.42$, $SD = 0.70$). Table 1 presents descriptive statistics for all main study variables. In line with hypotheses and paralleling results from Study 1, there was a significant association between differences in emotion beliefs and main study variables. As shown in Table 3, in line with hypotheses and similar to findings from Study 1, a greater difference between general and personal emotion beliefs in the direction of more malleable views of emotion in general compared to one's own emotions was positively correlated with higher levels of depression, stress, and negative affect. In line with predictions, a greater difference between general and personal emotion malleability beliefs was negatively associated with levels of positive affect (see Table 3). In line with hypotheses, a greater bias toward viewing one's own emotions as less malleable than emotions in general was associated with higher use of rumination. Contrary to hypotheses and findings in Study 1, there was not a significant association with the use of cognitive reappraisal. Results for when personal emotion malleability beliefs were the independent variable in analyses are presented in the Supplemental Material.

1-Month Follow-Up Analyses

We next examined these relationships longitudinally. Consistent with hypotheses, holding more malleable views of emotions in general compared to one's own emotions was positively correlated with greater depression at 1-month follow-up, $r(140) = .300$, $p = .005$, as well as higher levels of stress, $r(140) = .425$, $p = .002$, and negative affect, $r(142) = .315$, $p = .002$, at the 1-month follow-up. In terms of emotion regulation, similarly, holding more malleable views of emotion in general compared to one's own emotion was associated with the greater use of rumination as assessed at the 1-month follow-up, $r(140) = .241$, $p = .010$. Next, we conducted a series of follow-up linear regressions with the baseline version of the dependent variable as an additional predictor (e.g., baseline depression when depression at 1 month was the dependent variable). These analyses found that differences in baseline emotion beliefs remained a significant predictor of depression at 1 month, $B = .14$, $t(82) = 2.11$, $p = .038$. The relationships between baseline differences in emotion malleability beliefs and stress ($p = .984$), trait negative affect ($p = .167$), and rumination ($p = .874$) did not remain significant. This shows that differences in emotion beliefs relate to depression both concurrently and longitudinally above and beyond baseline levels of depression.

Table 3
Correlations Between Differences in Emotion Beliefs, Clinical Symptoms, and Emotion Regulation Assessed at Baseline and 1-Month Follow-Up in Study 2

Outcome variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. ITES general emotion beliefs	—	.77**	.14	-.17	-.02	.26*	-.05	.05	-.04	-.20	-.17	.23*	-.13	.28**	-.10
2. ITES personal emotion beliefs		—	-.53**	-.33**	-.26*	.29**	-.21*	.14	-.19	-.38**	-.46**	.26**	-.29**	.33**	-.31**
3. ITES baseline discrepancy			—	.21*	.29**	-.10	.28**	.23**	-.20*	.30**	.43**	-.10	.24**	-.15	.32**
4. BDI baseline score				—	.69**	-.33**	.46**	.65**	-.32**	.77**	.61**	-.34**	.56**	-.42**	.61**
5. PSS baseline score					—	-.24**	.56**	.61**	-.33**	.62**	.76**	-.28**	.59**	-.22*	.57**
6. ERQ baseline reappraisal score						—	-.13**	-.20*	.33**	-.34**	-.36**	.52**	-.19	.29**	-.26*
7. Brooding RRS trait score							—	.51**	-.18*	.49**	.62**	-.07	.72**	-.18	.59**
8. PANAS NA trait score								—	-.14	.68**	.58**	.30**	.56**	-.31**	.64**
9. PANAS PA trait score									—	-.32**	-.26	.34**	-.15	.44**	-.23*
10. IM BDI total score										—	.75**	-.38**	.66**	-.48**	.78**
11. IM PSS total score											—	-.28	.73**	-.48**	.68**
12. IM ERQ total score												—	-.21*	.47**	-.27**
13. IM RRS total score													—	-.23*	.69*
14. IM PANAS PA score														—	-.31**
15. IM PANAS NA score															—

Note. The ITES baseline discrepancy variable refers to the general–personal emotion malleability beliefs at baseline. Higher scores on this discrepancy variable refer to stronger beliefs that emotions in general are more malleable compared to one's own emotions. ITES = Implicit Theories of Emotion Scale; BDI = Beck Depression Inventory; PSS = Perceived Stress Scale; ERQ = Emotion Regulation Questionnaire; RRS = Ruminative Response Scale; PANAS = Positive and Negative Affect Schedule; PA = positive affect; NA = negative affect; IM = 1-month follow-up survey variables.
* $p < .05$. ** $p < .001$.

Exploratory Analyses

Next, a series of analyses was conducted to examine whether significant relationships between baseline emotion belief differences and clinical symptoms and affect at 1 month occurred via differences in emotion regulation. In the first series of analyses, baseline differences in general and personal emotion beliefs were the independent variables, and 1-month clinical symptoms, negative affect, and rumination were the dependent variables given significant relationships between these variables as assessed at 1 month with baseline emotion belief discrepancies. In these analyses, when clinical symptoms or negative affect was the dependent variable, differences in reported engagement in emotion regulation (specifically rumination) as assessed during the 1-month follow-up were the mediator variable. Likewise, when rumination as assessed at 1 month was the dependent variable in analyses, the level of emotional distress (e.g., depression, stress, or negative affect) as assessed at the time of the 1-month follow-up served as the variable linking these two constructs.

First, analyses were conducted to examine whether differences in emotional distress assessed as part of the 1-month follow-up accounted for the relationship between baseline emotion belief discrepancy and rumination at the 1-month follow-up (see Table 4 for all 95% CIs of the indirect paths). There was a significant indirect path linking baseline emotion belief discrepancies to rumination at 1 month via depression as assessed at the 1-month follow-up (indirect path: effect = 1.55, bootstrap SE = 0.70). More of a discrepancy between general and personal emotion malleability beliefs was associated with higher depression as assessed at the 1-month survey (β = .30, SE = 2.31, p = .005), which in turn was associated with higher rumination at the 1-month follow-up (β = .66, SE = 0.02, p < .001). The indirect

Table 4

Study 2 Significant Indirect Effects From Emotion Belief Differences at Baseline to Clinical Symptoms and Emotion Regulation at 1-Month Follow-Up

Dependent variable and mediator	95% CI
Rumination	
Via stress	[.38, 3.33] ^a
Via depression	[.24, 2.98] ^a
Via negative affect	[.53, 2.47] ^a
Via positive affect	[−.23, .62]
Reappraisal	
Via stress	[−1.89, 1.45]
Via depression	[3.47, −.26]
Via negative affect	[−2.35, −.15] ^a
Via positive affect	[−2.31, .62]
Stress	
Via rumination	[−1.20, 3.07]
Via reappraisal	[−1.45, 2.43]
Depression	
Via rumination	[−.34, 6.85]
Via reappraisal	[−2.06, 2.35]
Negative affect	
Via rumination	[.37, 3.01] ^a
Via reappraisal	[−2.70, .92]
Positive affect	
Via rumination	[−1.66, .03]
Via reappraisal	[−1.85, .76]

Note. CI = confidence interval.

^a Denotes a significant indirect path because the 95% CI does not include zero.

path in this model did not remain significant when baseline rumination was included as a covariate (indirect path 95% CI [−0.42, 1.28]). A similar pattern emerged for negative affect, in that more of a discrepancy between general and personal emotion beliefs at baseline was associated with more rumination at 1-month follow-up via differences in negative affect (see Table 4; indirect path: effect = 1.43, bootstrap SE = 0.49). Specifically, more of a difference in emotion beliefs was associated with more negative affect (β = .32, SE = 0.91, p < .002), which was then associated with more rumination at 1-month follow-up (β = .69, SE = 0.05, p < .001). The model was not significant when baseline rumination was included as a covariate (indirect path 95% CI [−0.25, 0.89]).

The relationship between baseline emotion belief differences and rumination at 1 month was accounted for by stress as assessed at 1-month follow-up (indirect path: effect = 2.06, bootstrap SE = 0.75; see Figure 1). Specifically, a greater difference between general and personal emotion beliefs at baseline was positively associated with higher stress at 1-month follow-up (β = .43, SE = .92, p = .002), and higher stress then related to higher use of rumination as assessed at follow-up (standardized β = .73, SE = 0.56, p < .001; see Figure 1). This model remained significant when baseline rumination was entered as a covariate (indirect effect, indirect path: effect = 0.74, bootstrap SE = 0.32, 95% CI [0.03, 1.31]).

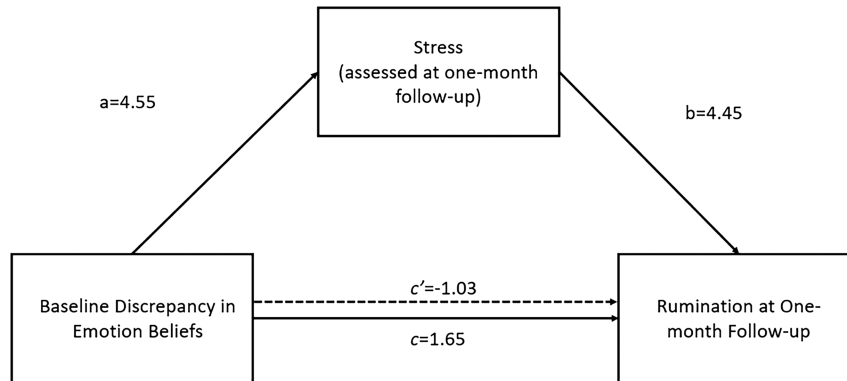
Next, a series of analyses examined whether differential engagement in emotion regulation, specifically rumination (as assessed at 1 month), accounted for the relationships between baseline emotion belief discrepancy and levels of emotional distress measured at 1 month (see Table 4). Initially, there was not a significant indirect path via rumination linking emotion belief differences to stress at 1 month because the 95% CI of the indirect effect included zero. However, there was a significant indirect path linking a greater difference in emotion beliefs at baseline to stress at 1 month via the use of rumination when the baseline level of stress was included as a covariate (effect = −1.32, bootstrap SE = 0.74, indirect path 95% CI [−3.07, −.18]). There was a significant indirect path linking baseline emotion belief differences to negative affect at 1 month via the differential use of rumination (as assessed at 1 month; effect = 1.56, bootstrap SE = 0.67; see Table 4). Specifically, stronger beliefs that emotions in general were more changeable than one's own emotions were associated with higher use of rumination as measured at the 1-month follow-up (β = .24, SE = 0.63, p = .010), and the greater use of rumination was associated with higher levels of negative affect at 1 month (β = .69, SE = .10, p < .001). However, the indirect path in the model with negative affect did not reach significance when baseline negative affect was included as a covariate (indirect effect 95% CI [−.36, 1.12]), likely due to baseline and 1-month negative affect being highly correlated, $r(141)$ = .64, p < .001. The model with depression at 1 month as the dependent variable did not reach significance (indirect path 95% CI [−.039, 6.53]).

Emotion Belief Differences and Depression Status

Next, on an exploratory basis, we examined whether the degree of the discrepancy between general and personal emotion beliefs varied based on depression status as determined by a semistructured clinical interview. In line with hypotheses, the nature of the discrepancy between general and personal emotion beliefs varied by depression status, $F(2, 140)$ = 3.56, p = .031, η_p^2 = .049. Individuals with no history or a past history of MDD (M = −0.18, SD = 0.50) did not

Figure 1

The Indirect Path Linking Emotion Belief Differences and 1-Month Rumination via Stress in Study 2



differ from each other in terms of emotion belief differences ($ps < .114$). However, individuals in a current depressive episode were significantly different from individuals with no history of depression (Tukey's $p = .043$). Specifically, individuals with no history of depression had more of a bias to viewing their own emotions as more malleable compared to emotions in general ($M = -.39$, $SD = .69$), whereas individuals in a current depressive episode did not show this bias ($M = 0.00$, $SD = .51$). Of note, the distribution of emotion belief discrepancy variable in the past MDD group was not normally distributed based on a Shapiro–Wilk test ($p = .001$). However, the distribution of this variable in both the MDD and no history of MDD groups (Shapiro–Wilk test $ps > .05$) was normally distributed and met the assumptions for an analysis of variance.

We next examined whether the correlations found between differences in general and personal emotion beliefs and clinical symptoms, emotion regulation, and affect varied based on depression status. Bivariate correlations within each group (no MDD, past MDD, and current MDD) found that a great discrepancy in emotion beliefs was positively correlated with higher depression, $r(43) = .37$, $p = .032$; higher stress (although this relationship was marginally significant), $r(43) = .28$, $p = .072$; and lower trait positive affect, $r(43) = -.38$, $p = .010$. For individuals with current MDD, a greater difference in emotion beliefs was associated with higher levels of trait negative affect, $r(16) = .54$, $p = .027$, although given the small sample size in the current MDD group, this exploratory analysis should be interpreted with caution. These correlations between emotion belief discrepancies and levels of emotional distress were not significant in the past MDD and no MDD groups ($ps > .082$). Among individuals with past MDD, but not current or no history of MDD, a greater difference in emotion malleability beliefs was positively correlated with the use of rumination, $r(78) = .26$, $p = .021$. There were no significant differences in the trait use of cognitive reappraisal based on depression status ($ps > .091$).

Discussion

The present study examined the role of discrepancies between beliefs about the malleability of emotion in general and one's own emotions in clinical symptoms, negative affect, and emotion regulation both concurrently and across time. Across Studies 1 and 2, more of a discrepancy between general and personal emotion beliefs

in the direction of holding more malleable views of emotion in general compared to personal emotion was associated with higher levels of depression, stress, and rumination and lower levels of positive affect. Study 1 found that less of a discrepancy between general and personal emotion beliefs was associated with lower levels of trait cognitive reappraisal; Study 2 found that a greater difference in emotion beliefs was associated with higher levels of negative affect.

Longitudinally, in Study 1, more of a discrepancy in emotion beliefs was negatively associated with the use of cognitive reappraisal at 1 month. Study 2 found that a greater difference between general and personal emotion malleability beliefs in the direction of holding more malleable views of emotions in general compared to one's own emotions was positively correlated with greater depression, rumination, and negative affect at 1-month follow-up. Results from analyses in Study 2 revealed a consistent pattern linking differences in emotion malleability beliefs at baseline to the use of rumination assessed longitudinally via the pathway of levels of psychological distress. When levels of clinical symptoms at 1 month were the dependent variables, there was a significant indirect path linking baseline emotion belief differences to negative affect via the differential use of rumination in that a greater difference in emotion malleability beliefs was associated with more negative affect, but not clinical symptoms, at 1 month via differences in the use of rumination.

The present studies' analyses examining the pathways linking discrepancies in emotion beliefs to emotional distress and emotion regulation extended past work that examined the pathways linking differences in emotion beliefs to emotional distress and emotion regulation. While past work has found that emotion malleability beliefs link to emotional distress via differential emotion regulation, such as cognitive reappraisal (E. T. Kneeland & Dovidio, 2020; Tamir et al., 2007), results from Study 2 provide support that emotion beliefs also can be associated with differences in emotion regulation, specifically the use of rumination, via levels of emotional distress. This pathway linking emotion beliefs to emotion regulation through differences in levels of emotional distress could reflect that individuals who hold more malleable views of emotion in general compared to their own emotions then feel higher psychological distress as a result, and this inclines them to rely on more avoidant strategies, such as rumination, across time as a way to cope with this elevated emotional distress.

In Study 2, we also examined whether differences in emotion beliefs varied by depression status (no MDD, past MDD, or current MDD). Results showed that individuals in a current depressive episode had less of a view that their own emotions were more malleable compared to emotions in general. Contrastingly, individuals with past depression or no history of depression had more of a bias toward viewing their own emotions as more changeable compared to emotions in general. These findings are in line with past work on negative biases in depressive cognition (Joormann et al., 2010) and extend this research into the domain of emotion beliefs to show that individuals do not hold a positive bias in their emotion beliefs when in a current depressive episode. These findings are important because a stronger view that their own emotions are more changeable than emotions in general could incline individuals to take a more active emotion regulation stance when faced with emotional upset. Given the central role of emotion regulation in conceptualizations of clinical depression (Joormann & Vanderlind, 2014), identifying factors that relate to and could potentially influence active emotion regulation represents a fruitful avenue for future research.

When we examined whether relationships between emotion belief differences and main study outcomes differed based on depression status, among individuals with no history of MDD, more of a difference in emotion beliefs was associated with greater depression and less positive affect, while this relationship was not significant in the remitted MDD or current MDD groups. This could reflect that there was a greater range in levels of depression and positive affect in the no MDD group, allowing for these significant relationships to emerge only in that group. We also found that only individuals with current MDD had a positive relationship between more of a difference in emotion beliefs and greater negative affect. This finding could show that within the context of a current depressive episode, viewing one's own emotions as less changeable than emotion in general exacerbates feelings of negative affect. This finding is important given that individuals with current depression struggle to regulate negative affect (Joormann & Stanton, 2016). Finally, only individuals with remitted MDD showed a significant association between more of a difference in emotion malleability beliefs and higher rumination. This finding could reflect that when individuals have had a past experience with depression and have the belief that their own emotions are less malleable compared to emotions in general, this stance could incline them to engage in repetitive thinking, such as rumination, to try to understand emotional distress.

The present study has several limitations that can guide future research. While the longitudinal nature of both studies represents a strength, both studies consist of self-report correlational data, and therefore the issue of causality remains. Future experimental work that directly manipulates the discrepancy about the malleability of general versus personal emotion beliefs could clarify whether a greater difference in general versus personal emotion malleability beliefs directly influences emotional distress and emotion regulation. If such a discordance in emotion beliefs is causal in influencing psychological distress and how individuals manage this distress, then interventions that reduce this discrepancy between believing that emotions in general are more malleable than your own emotions could be a target for intervention. Importantly, beliefs about the malleability of emotions and related constructs like anxiety and depression can be changed through brief experimental or single-session interventions (E. T. Kneeland, Nolen-Hoeksema, et al., 2016a, 2016b; Schleider & Weisz, 2018; Zimmermann & Papa, 2020).

In addition, Study 2 did not include consistent attention checks across the waves of data collection that were included in the present study's analyzed sample. Future research should use attention checks to better ascertain study adherence. In addition, in our analyses that examined whether variables such as rumination or depression symptoms were potential indirect paths linking baseline emotion belief discrepancies to the dependent variables of emotion regulation and clinical symptoms, these variables were also assessed as part of the 1-month follow-up. This limits the ability to speak to true mediation in the present investigation. Future research should utilize multiple time points to further explore mediation in relation to this research question.

Relatedly, given that depression status was assessed at the same time point as the assessment of individuals' emotion malleability beliefs, an area of future research could focus on what leads to a potentially harmful discrepancy in general versus personal emotion beliefs and how this can confer risk for and maintain clinical depression. It could be that increased and prolonged experiences with depression lead to a stronger belief that emotions in general are more malleable than one's own emotions; in fact, evidence suggests that certain types of depressive cognition can persist even when depression has remitted (Gotlib & Joormann, 2010). Or conversely, it could be that a greater discrepancy between general and personal emotion beliefs then increases risk for and maintains depression. Overall, examining longitudinally how a difference between general and personal emotion beliefs develops and interplays with depression status is worth future investigation.

Finally, future research should examine whether discrepancies in other types of emotion beliefs similarly relate to emotion regulation, affect, and clinical symptoms. Individuals can hold a range of beliefs about aspects of emotion such as the degree to which their own emotions are unique, last a long time, or serve a helpful function (Ford & Gross, 2019; Veilleux et al., 2015). Examining discrepancies between general and personal emotion beliefs in these domains represents a viable area for future research. For example, it could be that believing that your own emotions are more specific to you but others feel emotions more similarly could heighten feelings of loneliness and depression. Additionally, related features of emotion or of the individual, such as intensity of affect, could moderate the relationships found in the present study. For example, work by E. Kneeland et al. (2025) found that the relationship between emotion malleability beliefs and suicidal ideation varied based on the intensity of negative affect. Overall, future research could examine other factors that affect the relationship between emotion belief discrepancies and key mental health outcomes.

Constraints on Generality

Study 1 used a sample of individuals who were not prescreened for a specific level of emotional distress and were current undergraduate students at a small liberal arts college. Study 2 included individuals from surrounding communities in Western Massachusetts. Therefore, it is an open empirical question how the studies' results could generalize to individuals in other geographic locations within and outside of the United States. In addition, White/Caucasian was the most common race/ethnicity in both studies. It remains an important area for future research to examine the role of culture in the development of emotion beliefs and the discrepancy between these beliefs. Future work could examine how cultural norms related to individuals'

identities influence the relationships between discrepancy in emotion beliefs, clinical symptoms, and emotion regulation.

Conclusion

In conclusion, the current investigation addressed a current gap in the emotion malleability beliefs literature to clarify that a stronger belief that emotions in general were more malleable compared to one's own emotions was associated with higher levels of psychological distress and differences in emotion regulation both concurrently and across time. The present studies also clarified potential pathways in these relationships and how discrepancies in emotion malleability beliefs vary and relate to outcomes differently based on depression status. Future work could clarify causal mechanisms in how these beliefs relate to these important outcomes, developmental pathways that lead to this difference in emotion beliefs, and ultimately how we could intervene to reduce emotion belief discordance.

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